

Conditional Average Treatment Effects and Decision Making*

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Abstract

This paper develops a decision making empirical method to evaluate welfare programs accounting for heterogeneity of impacts. We find outcome predictive distributions for different subgroups of the population and use a characterization of second order stochastic dominance to give a policy recommendation conditional on covariates with minimal requirements on the social planner's utility function. Further, we can estimate quantile treatment effects within subgroups of the population. We apply this method to the Connecticut's Jobs First program and find subgroups for which the program did not maximize welfare even though some statistics may suggest the opposite.

JEL C11, D31, I38, J31

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1 Introduction

Studies evaluating social programs usually concentrate on overall population measures that miss the possible heterogeneity in the program impacts. Knowledge of the degree of heterogeneity is relevant to policy makers when extending the program to another jurisdiction or when reducing the number of beneficiaries since that may or may not bring potential welfare losses. This paper considers an empirical method to evaluate the heterogeneous impacts of social programs and how to map those into decisions using expected utility theory. Specifically, we find the probability distributions of potential outcomes for treated and controls for different subgroups of the population defined by covariates. Then we can determine whether a social planner with an increasing and concave utility function would prefer to assign a similar individual to the treatment or not by taking the expectation over the outcomes probability distributions. This can be done without having to specify the functional form of the planner's utility function. Moreover, with knowledge of the outcomes distributions we can also calculate quantile treatment effects within the groups. Thus the method gives not only information on the heterogeneity across groups but also within.

The problem can arise in the following situation, before implementing the social program to a larger population, a pilot phase is run in order to get insight about its benefits. Using the data collected in that phase, the social planner has to decide whether to implement the program or not on a larger sample, also the planner could decide to restrict the application of the program to a smaller population. The classical approach to answer this question is to find an overall treatment effect of the program and if it is statistically significant different from zero and positive, the program is likely to be preserved or extended. Here we propose a method to make inference on the treatment effect conditional on covariates even if the sample is very small within the subgroups and how to use that information in a decision making process.

A few papers¹ have considered the program evaluation problem as a decision problem:

¹Imbens and Wooldridge [2009] describe how the literature has concentrated on the overall treatment effect and argue that this would only make sense if we were to mandate exposure to the treatment to either everyone in the population or to no one, which in practice that is not usually the case.

by proposing a measure of the value of covariate information and how the higher variation in the treatment response as a function of covariates, the higher that value (Manski [2001], Manski [2004]); by looking at predictive outcome distributions for treated and controls and then using first order stochastic dominance to rank those distributions (Dehejia [2005]); and by extending and analyzing Manski’s approach to what the rules of assignment converge to in distribution when a loss function evaluates the decision rule and its interaction with risk rather than pure statistical rules (Hirano and Porter [2009]). On the other hand, another rather small branch of the literature has looked into the empirical problems to measure the heterogeneity of treatment effects (Abrevaya et al. [2012], Bitler et al. [2006], Bitler et al. [2010]); and by considering the theoretical issues on the hierarchical Bayesian models for evaluation programs when there is heterogeneity and the use of an empirical Bayesian approach (Chamberlain [2011]). The method we develop in this paper connects these two branches of the literature in order to be able to evaluate using expected utility theory whether assignment to treatment is beneficial in a subgroup of the population defined by covariates or not. We use a model that captures correlations across the subgroups of the population through a hierarchical Bayesian model and then second order stochastic dominance to map the within subgroups probability distributions of potential outcomes into expected utility theory.

At a broader scale, methods to estimate treatment effects can be classified as follows²: (i) regressions of the outcome on the covariates and a treatment dummy, (ii) matching on covariates, (iii) using the propensity score, (iv) combinations of those methods, and (v) Bayesian methods. This paper uses a hierarchical Bayesian model to get posterior distributions for key parameters and then use those to construct predictive distributions of outcomes. The use of Bayesian methods to estimate treatment effects dates back to Rubin [1978]³. After that paper, little has been done on estimating treatment effects using Bayesian techniques

²Imbens [2004]. Heckman [2010] gives an in-depth analysis of the state-of-the-art on the structural versus the reduced form approaches to do empirical work on program evaluation.

³He studies the estimation of causal effects by finding the predictive distribution of the outcomes that correspond to the treatments that are not available in the experiment. The idea is that the unobservable outcomes (because an individual can be assigned to either the treatment or to the control group but not both) are missing data and those values can be imputed by giving a prior distribution on the potential outcomes.

(Imbens [2004]), perhaps because of the intensive computational requirements.

The use of the covariates is crucial, however we do not assume any particular distribution on them. Rather we group the individuals with similar characteristics and estimate the treatment effect for each group. The groups could share more or less information depending on the unknown distribution of the covariates, thus we allow for correlation across the groups. This approach could be particularly useful when the number of covariates is much bigger than the number of observations, which is a typical situation in medical surveys for instance.⁴ The methods mentioned above can have some weaknesses in these cases. Imbens [2004] points out that when there are many covariates, non-Bayesian methods tend to include all of them without restrictions as if they were equally important in the model or exclude some of them. In general, estimation of treatment effects for small samples or for very restrictive subsets of covariates has been avoided because of the potential lack of statistical significance under classical approaches⁵.

We apply the method to the Connecticut’s Jobs First program. This arised as one of the welfare reform experiments in the U.S. caused by the elimination of the Aid to Families with Dependent Children (AFDC) program in 1996. The federal government required the states to replace AFDC with programs that had a time limit to participate and that would enhance job training. Thus we focus in this paper on the transitional program Jobs First implemented in Connecticut, specifically in Manchester and New Haven. The welfare experiment consisted in randomly assigning 4,803 people with at least one dependent children⁶ to either AFDC or to Jobs First. The outcome of interest is the average quarterly income defined as the average of earnings plus transfers in the form of food stamps over the first seven quarters after inception to the experiment. One of the main differences of Jobs First with respect to AFDC is that the former has a limit of time of cash assistance of 21 months plus 6 months if an extension is requested and approved. This feature coupled with the requirement of following a job training program instead of just general education programs, were expected

⁴As in Efron [2011].

⁵Graham and Hirano [2011] discuss the performance of different estimators when there are empty cells.

⁶Of these, 96% were women.

to make a difference in the level of the impact of these welfare programs.⁷ It is however unclear how the impacts of the program vary with covariates, and even less so how to make decisions on extensions of the program based on expected utility arguments. The impacts here are the differences between the outcome level if the individual was in the experimental group and the outcome level if she was in the control group.

Perhaps the two closest papers to ours are Bitler et al. [2006] and Dehejia [2005]. In the former they measured heterogeneous impacts of the Connecticut’s Jobs First program by looking at the differences in income between treated and controls at each different quantile of the income distribution. Dehejia [2005], using data from a randomized experiment, estimates posterior densities for the treatment effects and evaluates under different scenarios the maximization of welfare using the posterior densities. His contribution is that he models program evaluation as a decision problem. For example, in order to decide whether all the individuals should be assigned to the treatment. Another scenario is the one of a caseworker that has to decide if a person should be assigned to the treatment or to the control group. In order to carry out these decisions, he uses classical expected utility theory since the predictive distributions for the outcomes are known in a Bayesian framework. The main difference with respect to this paper is that we are also going to condition on covariates so that the heterogeneity problem overlaps with expected utility rankings.

Our results suggest that even though classical statistical measures for conditional treatment effects for only one subgroup in our main specification is negative, when using the expected utility approach there are two other subgroups for which exposure to the treatment does not maximize utility. This difference is due to the lack of statistical power of the classical approach and the lack of use of information across subgroups. There is also a high degree of heterogeneity in quantile treatment effects within the subgroups. A second specification in which pre-treatment covariates are discarded when forming the subgroups wipes out most of the heterogeneity.

We believe that it is of relevance to understand the impacts of social programs not just as an overall measure for the entire population but rather how large the heterogeneity of

⁷For more details on Jobs First see the MDRC’s Final Report (Adams-Ciardullo et al. [2002]).

these impacts is and how to make assignment decisions based on robust theoretical expected utility results that do not depend on particular functional forms but only minimal conditions. The rest of the paper is organized as follows. Section 2 presents the definitions and formal statements of the problem. Details on Jobs First and some data issues are explained in Section 3. Section 4 presents a hierarchical Bayesian model and a Gibbs sampler. Finally, in Section 5 we present the connection between predictive income distributions and expected utility theory as well as the main results. Section 6 concludes.

2 Definitions and Statement of the Problem

A social planner is trying to decide whether an individual should be assigned to a treatment or not to maximize an outcome. In this paper, the outcome is earnings plus transfers. Other possibilities are health quality, level of education, and in general, any outcome of a social program that can be quantified. Our social planner has data on N individuals. The data for each individual consists of individual characteristics, whether the individual received the treatment or not, and the outcome. In order to do the estimation, some individuals are randomly given the treatment. Each individual is characterized by a vector of covariates X_i of size $1 \times K$. Denote the fact of being exposed to the treatment by $T_i = 1$ and by $T_i = 0$ if the individual did not receive the treatment. The outcome will be denoted by

$$Y_i \equiv T_i \cdot Y_i(1) + (1 - T_i) \cdot Y_i(0).$$

The following assumption, known as the unconfoundedness assumption and first proposed by Rosenbaum and Rubin [1983], captures the idea that treatment assignment is exogenous and allows to take the expectations below,

$$(Y_i(0), Y_i(1)) \perp T_i | X_i.$$

This means that once you condition on X , treatment assignment is independent of the potential outcome. However, the realization of T does affect which outcome is observed. This assumption is also known as exogeneity.

We are interested in estimating the effect of being under the treatment for a given X . As one observes either $Y_i(0)$ or $Y_i(1)$ but not both at the same time for individual i , we always have some uncertainty in one of the two outcomes. We are interested in estimating the expectation of the difference of the outcomes had the individual been in both cases. We call this quantity the *average treatment effect* (ATE) and we define it as

$$\beta := E(Y(1) - Y(0)).$$

If we are interested in the treatment effect for individuals in the cell $X = x$, we define the *conditional average treatment effect* (CATE),

$$\beta(x) := E(Y_i(1) - Y_i(0) | X = x).$$

Note that this can be interpreted as an average treatment effect for a group of individuals with characteristics $X = x$. We can rewrite $\beta(x)$ as follows,

$$\begin{aligned} \beta(x) &= E(Y_i(1) - Y_i(0) | X = x) \\ &= E(Y_i(1) | X = x) - E(Y_i(0) | X = x) \\ &= E(Y_i | T = 1, X = x) - E(Y_i | T = 0, X = x) \end{aligned}$$

where $Y_i = T_i \cdot Y_i(1) + (1 - T_i) \cdot Y_i(0)$. The overall ATE can be obtained by taking the expectation with respect to X ,

$$\beta = E(\beta(x)).$$

This suggests that estimation for $\beta(x)$ consists of two parts, $\beta_1(x) = E(Y | T = 1, X = x)$ and $\beta_0(x) = E(Y | T = 0, X = x)$. A simple estimator for the CATE can be constructed using sample means,

$$\begin{aligned} \hat{\beta}_1(x) &= \frac{1}{N_1} \sum_{i: T_i=1, X=X_i} Y_i \\ \hat{\beta}_0(x) &= \frac{1}{N - N_1} \sum_{i: T_i=0, X=X_i} Y_i \end{aligned}$$

where N_1 is the number of individuals with covariate $X = x$ and under the treatment. An estimator for the CATE is the difference

$$\hat{\beta}(x) = \hat{\beta}_1(x) - \hat{\beta}_0(x). \tag{1}$$

It is a consistent estimator⁸ and its variance is

$$\begin{aligned} Var(\hat{\beta}(x)) &= Var(\hat{\beta}_1(x)) + Var(\hat{\beta}_0(x)) \\ &= \frac{1}{N_1}\sigma_1^2(x) + \frac{1}{N-N_1}\sigma_0^2(x), \end{aligned}$$

where $\sigma_1^2(x)$ and $\sigma_0^2(x)$ are the variance of the outcomes for the treated and the controls respectively when $X = x$. By replacing the variances with their sample analogs we have $\widehat{Var}(\hat{\beta}(x)) = \frac{1}{N_1}\hat{\sigma}_1^2(x) + \frac{1}{N-N_1}\hat{\sigma}_0^2(x)$.

Our problem is to find a rule to decide whether a new individual $N+1$ should be assigned to the treatment or not. Suppose that the social planner wants to maximize the population mean outcome of the treatment if she has an increasing and concave utility function. This is equivalent to maximize the outcome of each individual. That is, conditional on X , she wants to choose T_{N+1} such that

$$T_{N+1} = \arg \max_{t \in \{0,1\}} \{E(Y_{N+1}(t)|X_{N+1})\}$$

so that the population mean outcome is

$$E\left(\max_{t \in \{0,1\}} \{E(Y(t)|X)\}\right)$$

where the outside expectation is with respect to X (see Manski [2001]). Note that we would like to have the distribution of Y in order to compute the inner expectation. It is the Bayesian framework that allows one to construct predictive distributions on Y .

The maximization problem above is difficult to solve because for each individual we observe only one of the potential outcomes, then usually ATEs are estimated as an overall value for the whole sample (regression methods) or by groups of individuals with similar characteristics (propensity score methods, matching). As in Manski [2001] and in Manski [2004] it is

⁸By using the unconfoundedness assumption to get the third equality,

$$\begin{aligned} \text{p lim } \hat{\beta} &= E(Y_i|T=1, X=X_i) - E(Y_i|T=0, X=X_i) \\ &= E(Y_i(1)|T=1, X=X_i) - E(Y_i(0)|T=0, X=X_i) \\ &= E(Y_i(1)|X=X_i) - E(Y_i(0)|X=X_i) \\ &= \beta \end{aligned}$$

pointed out, usually the methods for ATEs estimation focus on population-wide estimators in order to be able to use the classical approaches of hypothesis testing that rely on large sample properties. Then researchers might not take into account the information of the covariates through the construction of subsamples of individuals with similar characteristics because that potentially diminishes statistical significance due to the small subsample sizes. Dehejia [2005] proposes a Bayesian estimation of the ATE and obtains predictive distributions for earnings for treatments and controls, this then can be combined with a stochastic dominance criterion to make assignment decision. The approach followed here is to use the information of the covariates by grouping the individuals according to their covariates. Specifically we use a characterization of second order stochastic dominance. If we have two random variables X and Z , $E(u(X)) \geq E(u(Z))$ for *any* increasing and concave function u if and only if X second order stochastically dominates Z . The latter statement can be evaluated using the predictive distributions obtained from the data.

Remember that X_i contains K different covariates. The social planner has some information to decide how to discretize each covariate. In the empirical application it is assumed that this criterion is exogenous to the problem of maximization of the expected value of the outcome and the level of discretization is left for future research. For example, if the covariate is age the social planner might discretize the variable into young, adults and seniors according to specific ranges for the covariate age. If the sample is big enough the level of discretization can be less coarse. A cell is each of the elements of the discretization over all the covariates. In the example, young, adults and seniors are the three cells obtained by the discretization. For identification purposes, we want that in each of the cells there is at least one individual under the treatment and at least one individual in the control group. That is, in each segment of each covariate there exists at least one pair $(X_i, Y_i(0))$ and at least one pair $(X_i, Y_i(1))$. Once we have the discretization for each covariate we can form subgroups of individuals according to all the possible combinations of the segments. This means that we take the cartesian product over the segments of the covariates. For example, if we have another covariate for education say, we can discretize it in three levels corresponding to different levels of education (low, medium, high) for some specific ranges in the covariate.

We can combine these cells with the three cells of the covariate age to get nine different subgroups⁹. In this way, every individual belongs to exactly one subgroup and only one and each subgroup contains at least one pair $(X_i, Y_i(0))$ and one pair $(X_i, Y_i(1))$.

One could estimate the CATEs by taking the difference of the sample means in each subgroup. But this might not give any meaningful results since the size of the subgroup could be very small. That would not be taking into account the correlation with other cells either. The assigning rule would now be to take the $(N + 1)$ -th individual, and according to her characteristics $X_{N+1} = x$, she would belong to one of the subgroups, and she would be assigned to the treatment if the random variable $Y|T = 1, X = x$ second order stochastically dominates $Y|T = 0, X = x$.

3 Connecticut's *Jobs First* Program

Connecticut's Jobs First was put in place just before the passage of the federal welfare reform through the Personal Responsibility and Work Opportunity Act (PRWORA) signed by President Bill Clinton in 1996. One of the main consequences of PRWORA was the substitution of AFDC with the Temporary Assistance for Needy Families (TANF) program. Jobs First was one of the very first programs with all the TANF main features to be implemented, and it is thus one of the few examples available to get an insight on the performance of TANF programs versus the previous AFDC format. The main differences between the two are that TANF-type programs have time limits and work requirements where as AFDC did not.¹⁰

The time limit consisted on a limit of 21 months of cash assistance, with the possibility of an extension of 6 months if the recipient complied with requirements for an exemption. AFDC did not have any time limit. TANF now allows for up to 60 months of cash assistance. We should keep in mind though that about two thirds of those in our sample who reached the time limit under Jobs First were approved the extension of 6 months. This is why we look at the quarterly average of income over the first seven quarters after inception to the program.

⁹One subgroup for example is the one that contains young people with high level of education.

¹⁰Adams-Ciardullo et al. [2002].

The second key feature of Jobs First is that the earnings from working are disregarded when calculating the food stamps benefits as long as these earnings were below the FPL¹¹. This was not the case under AFDC where the earnings disregarded were a function of the length of the time in the program. After 12 months under AFDC, \$90 were disregarded per month when calculating the food stamps benefits.

Other features of Jobs First are that the asset limit for cash assistance is \$3,000 whereas under AFDC was \$1,000. The benefit per child conceived while mother was receiving cash assistance was \$50 per month under Jobs First and about \$100 under AFDC. Jobs First was a state-wide program, however the data we use here come from only two locations: New Haven and Manchester which altogether account for about one fourth of the entire caseload. Among all these recipients, 96% were women. All recipients are eligible only if they have at least one child at the moment of inception to either program.

Estimates on the mean impact of Jobs First versus AFDC have been found to be \$453 on income¹² per quarter according to the Final Report (Adams-Ciardullo et al. [2002]). Other studies such as Bitler et al. [2006] have found a mean impact of \$294 per quarter over the first seven quarters in the program. Moreover, Bitler et al. [2006] found significant heterogeneous effects when looking at the impacts conditional on quantiles of the income distribution. These impacts range from 0 to \$800. The impacts increase with the quantile and then decrease towards the upper quantiles. This may be attributed to an opt-in effect as some individuals might lower their income in order to be eligible for the program. The type of heterogeneity they looked at is the difference between the cumulative distribution of income for the treated and for the controls at each different quantile.¹³ For example, at the q -th quantile, the difference $Y_{1,q} - Y_{0,q}$ is the quantile treatment effect, where $F_i(Y_{i,q}) = q$ is the cumulative distribution function of income Y for group i . These large variations in the impacts conditional on income quantiles is just one piece of the puzzle. We can further explore the impacts heterogeneity conditional on covariates.

¹¹Federal Poverty Line, which was \$1,138 per month for a family of three people in 1998.

¹²Income is defined as the sum of earnings from working and the food stamps benefits.

¹³The horizontal difference between the graphs of the cumulative distribution functions for any given number between 0 and 1.

The sample consists of 2,396 Jobs First recipients and 2,407 AFDC recipients, making a total of 4,803 families. In this paper we focus on four covariates: age, whether the individual was employed before inception to program, number of children, and high school degree.¹⁴ The partition of the sample consists of the cartesian product of the segments created in each variable as follows: the covariate age is segmented in two parts depending on whether the parent is at least 30 years old or not; a dummy variable for employment before the program; dummy variables for being pregnant or with one kid, with two kids, or with three or more kids; and whether the parent has a high school degree or not. This gives a total of $J = 24$ subgroups and a total of 4,463 observations due to missing values in the high school degree covariate. Table 1 describes the groups and Figure 1 shows histograms of the data by group and by treated and controls. As explained before, our outcome variable is the quarterly average of earnings plus transfers over the first seven quarters after inception to the program. 27% of the sample has zero earnings, 5.5% no transfers, and 1.2% none. This produces a mass point over zero on the empirical income distributions, an issue we further discuss in the next section.

The particular choice of the subgroups is certainly exogenous in the model. This however has been done using covariates that the Final Report uses as explanatory variables in their estimation of impacts. Bitler et al. [2010] use the same covariates as well to diagnose subgroup heterogeneity in Jobs First. In general this is a good approach to form the subgroups because it is realistic to think that a social planner extending or constraining the program to a different population will be able to sustain her criteria if they are based on such covariates because age can determine your expertise in your job and the number of kids your actual economic needs. Having being previously employed is also used as a criterion for welfare aid, such as in unemployment benefits. Even though we have data on race, we believe its use in reality could bring problems as it could be taken as racial profiling. In the robustness section we discuss two other different specifications for the subgroups, using the race information and using only non-historical covariates.

¹⁴Later in the paper we present the results when race is used for making the subgroups, however, this may not happen in reality as that may be considered racial profiling.

As a benchmark for the results, Table 3 reports the estimates for the CATEs using the estimator (1). We can see the tremendous amount of variation between the subgroups' CATEs. The goal is to find which subgroups have a high value for their CATE, not only by taking the within subgroup sample means because of the lack of statistical significance we could get in some of the subgroups, but from a decision making standpoint. The last column in Table 3 shows the t-statistic for the null hypothesis that the CATE is zero¹⁵ and how in most subgroups the null hypothesis cannot be strongly rejected. This paper proposes how to calculate predictive probability distributions for income and how to use them in an expected utility maximization setting.

4 Estimation and Results

Once the subgroups are formed, the sample can be described by pairs (T_{ij}, Y_{ij}) where i represents the individual observation and j is the index for different subgroups in the sample. Each subgroup can have different number of observations, so $i = 1, \dots, n_j$ and $j = 1, \dots, J$ where J is the number of subgroups. A first natural approach is to estimate CATEs by taking the difference in sample means. Table 3 shows the results. There are two observations to make, first, statistical significance of these differences as measured by the t-statistic is rarely found, and second, among the CATEs that exhibit certain degree of significance it is evident that the size of the program impacts substantially differs from one subgroup to the other. The former is just a consequence of the lack of statistical power of this simple estimator on small sample sizes. The second suggests that in effect, the impact of the program is not constant across subgroups. A different approach, which also relies on asymptotics, is to create dummy variables for each subgroup and then run an OLS on these dummies and their interactions with the treatment indicator variable. The results are shown in Table 2. Once again, it is rare to come across with a statistically significant coefficient.

¹⁵These are calculated to test whether the two means are statistically significantly different. The statistic is $t = \frac{\bar{Y}(1) - \bar{Y}(0)}{\sqrt{\frac{(n_1 - 1)S_1^2 + (n_0 - 1)S_0^2}{n_1 + n_0 - 2} \sqrt{\frac{1}{n_1} + \frac{1}{n_0}}}}$ where S_i^2 are the sample variances for the treated and the controls. See Dehejia [2008] for more details on rules-of-thumb for inference of treatment effects.

4.1 A Hierarchical Bayesian Model

To overcome issues of estimators that rely on asymptotics we use a Bayesian approach instead. We assume that each realization of the random variable Y_{ij} is an independent draw from a normal distribution,

$$y_{ij} \sim N(\mu_j, \tau_{\mu j}).$$

So each group has its own mean μ_j and own precision $\tau_{\mu j}$ and from now on, the second parameter in a normal distribution will refer to the precision not the variance, but we want to capture the fact that these means are not independent which reflects the fact that there is some correlation across the subgroups. That is, there is an underlying common distribution for all the subgroups from which we draw these means,

$$\mu_j \sim N(\mu, h_1),$$

where μ is the overall outcome mean. If we independently draw from that distribution we have that the group means $\mu_j|\mu$ are conditionally independent. However, without conditioning on μ , the subgroup means are correlated. Since we do not know how the subgroup means μ_j are allocated, we give a prior on the precision h_1 that captures whether the subgroups share more or less information. The prior to be adopted is a gamma distribution with most of its mass close to the origin which is equivalent to have a diffuse prior on μ_j .¹⁶ Here, the gamma distributions will be denoted as $G(a, b)$ with mean a/b and variance a/b^2 . Suppose that the subgroups share much information, then the subgroup means μ_j will be similar to each other. On the other hand, if the subgroups do not share much information, we expect the subgroup means μ_j to be very different and the posterior distribution should be using little information from μ to estimate the subgroup means.

The normality assumption for the outcome is not ideal since there is a probability mass point on the left of the income distribution at zero as explained in the previous section. Figure 1 shows the histograms of the treated and the control subsamples for the outcome variable. The censoring at the left of the distribution occurs because of the people in the

¹⁶See details in the Appendix

sample that are unemployed and did not receive transfers in the form of food stamps benefits. For this reason we choose a Tobit likelihood for the model.¹⁷ First we present the model as if there was no mass point on zero and then present the full model.

Assume the prior on μ and on ν ,

$$\mu \sim N(\nu, h_2) \tag{2}$$

$$\nu \sim N(0, h_3)$$

and a gamma density prior on $\tau_{\mu j}$ for each $j = 1, \dots, J$,

$$\tau_{\mu j} \sim G(\alpha, \beta).$$

The numbers h_3, α and β are fixed. Below we give a prior on h_1 and h_2 . This completes the model, a hierarchical Bayesian model.

The joint posterior distribution is proportional to

$$\begin{aligned} p(\mu_1, \dots, \mu_J, \mu, \tau_{\mu 1}, \dots, \tau_{\mu J} | y_{ij}, h_1, h_2, h_3, \alpha, \beta) &\propto \left(\prod_{i,j} \tau_{\mu j}^{1/2} \exp \left(-\frac{1}{2} \tau_{\mu j} (y_{ij} - \mu_j)^2 \right) \right) \times \\ &\quad \left(\prod_{j=1}^J h_1^{1/2} \exp \left(-\frac{1}{2} h_1 (\mu_j - \mu)^2 \right) \right) \times \\ &\quad h_2^{1/2} \exp \left(-\frac{1}{2} h_2 \mu^2 \right) \times \\ &\quad \prod_{j=1}^J \tau_{\mu j}^{\alpha-1} \exp(-\beta \tau_{\mu j}) \end{aligned}$$

where each term of the product above is the kernel of the densities for $y_{ij}, \{\mu_j\}_{j=1,\dots,J}, \mu$ and $\{\tau_{\mu j}\}_{j=1,\dots,J}$ respectively. Figure 2 shows a graphical representation of this hierarchical Bayesian model.

The estimation is done using a Gibbs sampler. In general, this method produces, when the number of iterations is large enough, a random draw from some density $p(\theta|A)$, where θ is the random vector of interest and A is the set of fixed parameters. Assume that θ can be

¹⁷A third suitable likelihood is a mixture of a mass point on zero and a normal density on the $\log(\text{income})$ for the positive observations

decomposed in blocks of parameters $\theta = (\theta_1, \theta_2)$, so that $p(\theta|A)$ can be written as $p(\theta_1|\theta_2, A)$ and as $p(\theta_2|\theta_1, A)$ and it is possible to get random draws from these two densities. In this setting, a Gibbs sampler consists of getting a random draw $\theta_1^{(1)}$ from $p(\theta_1|\theta_2^{(0)}, A)$ using some initial value $\theta_2^{(0)}$ where the superscripts denote the number of the iteration, then getting a random draw $\theta_2^{(1)}$ from $p(\theta_2|\theta_1^{(1)}, A)$. Note that an initial value was needed in order to sample from $p(\theta_1|\theta_2, A)$, and also note that we update the values of θ_i in the next iteration. If the iteration is done N times, we get a random sample of size N of draws for θ_1 and for θ_2 . Under some general conditions, it can be shown that¹⁸

$$E(g(\theta)) \approx \frac{1}{N} \sum_{j=1}^N g(\theta_1^{(j)}, \theta_2^{(j)}).$$

for large N . Going back to the estimation of CATEs, it is of interest to write the joint distribution as a set of full conditional distributions. Because of the form of the joint distribution in this particular setting, we can write the full conditional distributions as¹⁹

$$\begin{aligned} \mu_j | \mu, \mu_{-j}, \tau_\mu, y_{ij}, h_1, h_2, \alpha, \beta &\sim N \left(\frac{\sum_{i=1}^{n_j} y_{ij} + h_1 \mu / \tau_{\mu j}}{n_j + h_1 / \tau_{\mu j}}, \tau_{\mu j} n_j + h_1 \right), \quad j = 1, \dots, J, \\ \mu | \{\mu_j\}_{j=1, \dots, J}, \{\tau_{\mu j}\}_{j=1, \dots, J}, y_{ij}, h_1, h_2, \alpha, \beta &\sim N \left(\frac{\sum_{j=1}^J \mu_j}{J + h_2 / h_1}, J h_1 + h_2 \right), \\ \tau_{\mu j} | \mu, \{\mu_j\}_{j=1, \dots, J}, \tau_{\mu -j}, y_{ij}, h_1, h_2, \alpha, \beta &\sim G \left(\alpha + n_j / 2, \beta + \frac{1}{2} \sum_{i \in j} (y_{ij} - \mu_j)^2 \right), \quad j = 1, \dots, J. \end{aligned}$$

These three densities can be used in a Gibbs sampler. We can see that the posterior mean of the subgroup means²⁰ $\mu_j|A$ is a weighted average of the outcomes within the subgroup and the overall outcome mean μ . The precision is a weighted sum of the precision of the income for subgroup j and the precision of μ_j . The overall mean μ is weighted by the precision of subgroup j . Also, the precision of the overall distribution is weighted by the precision of group j . Observe that if n_j increases (there are more individuals in subgroup j), the posterior precision $\tau_{\mu j} n_j + h_1$ increases, which is desirable.

¹⁸See Geweke [2005].

¹⁹See Appendix for details.

²⁰The letter A henceforth will denote all the parameters and data we are conditioning on at that particular step of the sampler.

From the posterior distribution of μ we see that its posterior mean is almost the average of the subgroup means, but the denominator contains the term h_2/h_1 . This suggests that h_2/h_1 should be small compared to J , the number of subgroups, in order to let the data dominate and not the initial values of the parameters. Then, in the specification of the priors we will require $h_2 < h_1$. As shown in the Appendix, the posterior distributions for these hyperparameters do not explicitly depend on the value of the observations. The posterior precision suggests that when the number of subgroups increases, the precision is higher and that h_2 , the precision in the prior for μ , should be very small to have an uninformative prior on μ .

The third posterior distribution is the collection of posteriors for the subgroup precisions. Observe that these hyperparameters do not depend on h_1 or h_2 in the conditional density, but in the number of total observations and the overall sum of the squares of the deviations from the subgroup means. The parameters α and β should take on values that do not dominate the data. Let $\epsilon_j \equiv \frac{1}{2} \sum_{i \in j} (y_{ij} - \mu_j)^2$. The mean of τ_{μ_j} is clearly not dominated by the parameters if α is much smaller than half the size of the smallest subgroup, and if β is much smaller than ϵ . However, when there is not much variation in the outcomes within the subgroup, i.e. when $\epsilon \rightarrow 0$, the expected value should be large enough to reflect the fact that the subgroup variance is close to 0. Therefore $(\alpha + n_j/2)/\beta$ cannot be too small. At the same time, the variance for this conditional density is $(\alpha + n_j/2)/(\beta + \epsilon)^2$ and it should be relatively larger in the $\epsilon \rightarrow 0$ case since there is not much information to accurately infer the subgroup variance. A good compromise on these parameter values is as shown in Table 5.

Finally we introduce a prior on h_1 , the precision of the distribution of μ_j ,

$$h_1 \sim G(\gamma, \gamma),$$

and a similar prior on h_2 . Remember that the posterior distribution for μ suggests that $h_2 < h_1$ is a good selection for the initial value of these parameters. The posterior distribution for $h_1|A$ where A represents the rest of the parameters, is $h_1|A \sim G(\gamma + J/2, \gamma + \frac{1}{2} \sum_j (\mu_j - \mu)^2)$.²¹ Note that if the subgroup means μ_j are very far from each other then the posterior mean for h_1 is close to zero, which implies a large variance in the distribution of the subgroup means.

²¹See Appendix for details.

The posterior density for h_2 is $h_2|A \sim G(\gamma + 1/2, \gamma + \frac{1}{2}\mu^2)$ so the dispersion of the subgroup means depends on their overall mean and initial parameters, however the influence of the latter is negligible compared to the size of the other arguments.

The posterior densities above form a Gibbs sampler. Given some initial values $(\mu_j, \mu, \tau_{\mu j})$ for $j = 1, \dots, J$, the posterior distribution for μ_j is known and we can get a draw from that distribution and update those values to draw a value for μ , then we use that value to draw values for the $\tau_{\mu j}$'s from there posterior distribution. We repeat this process until getting convergence for all the posterior distributions. The estimation is made separately for the treated and the controls. The CATE for subgroup j is defined as $\mu_j^{treated} - \mu_j^{control}$. The link between $\mu_j^{treated}$ and $\mu_j^{control}$ is the prior from which the overall mean for the treated and the overall mean for the controls are drawn as in (2). The specification also allows for heteroskedasticity across subgroups and different variance for the overall mean of treated and controls.

As pointed out above, the data show evidence of censoring close to the zero income point. To alleviate this problem we include, on top of the Gibbs sampler explained above, a data augmentation step for Tobit models first proposed by Chib [1992]. Once the mass point at the extreme left of the distribution has been replaced with hypothetical negative observations, the Gibbs sampler above is applied in the exact same way as before²².

4.2 Posterior CATEs

After an amount of iterations of the Gibbs sampler²³ and a burn-in phase²⁴ the histograms for all the parameters' posterior distributions show convergence in distribution, see Figure 3 which shows the time series of random draws for selected parameters from the treated subsample. We can now calculate posterior means for the parameters, in particular we are interested in the posterior means for μ_j and the $\tau_{\mu j}$. Table 7 presents the results by subgroup. The first five columns from the first part of that table show the posterior means and

²²See Appendix for details

²³2,000 iterations for the outer loop and 2,000 iterations for the data augmentation step.

²⁴The first 1,000 random draws from the Gibbs sampler are discarded to avoid any influence from the initial parameters of the priors in the final results.

the posterior standard deviations for the treated. The last four columns do the same for the controls. The bottom part of the table shows posterior means for the hyperparameters. Notice that even though the posterior mean values for the precisions $\tau_{\mu j}$ are in the order of 10^{-7} , the implied subgroup standard deviations are in the order of 10^3 dollars, which is in the same order of magnitude as our data on outcomes²⁵.

When looking at the distributions of predicted income²⁶, see Figure 4, we can see how the mass point near zero has been spread out over negative values of income. This is an essential step to tackle the censoring problem in the data. With posterior densities of the subgroup means in hand we can calculate posterior densities for the CATEs for each subgroup. For every draw of the joint posterior distribution of parameters we find the predicted income for treated and for controls and their difference, this is the conditional average treatment effect posterior distribution. Figure 5 and the last two columns of Table 4 show these results. Figure 5 plots the kernel density estimates for the distribution of the posterior of CATEs. Observe that the sign of the treatment effect does not change in the posterior means with respect to the sample means results. However, the level of the effects slightly changes in most cases. We can also see from the posterior densities of the CATEs the heterogeneity of the impacts within each of the subgroups. For example, subgroups 9, 10, 12, and 15 have a lot of heterogeneity if measured by the spread of the distribution. The first three correspond to people who were previously employed and no high school degree, whereas the last one corresponds to people with high school degree and three or more children.

5 Welfare

We can form predictive distributions for income in the following way.

1. Get a random draw from the posterior distributions $p(\mu_j|A)$ and $p(\tau_{\mu j}|A)$.
2. Get a random draw from

$$p(y_{ij}|A).$$

²⁵We further discuss sensitivity to different initial parameter values in the section on robustness checks.

²⁶The next section explains how to construct predictive income distributions.

Where A represents all the other parameters and data respectively updated at each step.

The social planner is concerned with the optimal choice of subgroups to be assigned to or exempted from the treatment in order to maximize welfare. Using the definition of second order stochastic dominance we can get a natural welfare interpretation of the relationship between the posterior predictive distributions within each group.

Define $G_X(a)$ as the integral of the cumulative distribution function,

$$G_X(a) = \int^a F_X(t)dt$$

for a random variable X with density dF_X and a real number a . Using this notation, a random variable X *second order stochastically dominates* (SOSD) a random variable Y if

$$G_X(a) \leq G_Y(a) \tag{3}$$

for all a in the support of the distributions, where G_i is defined as above.

A well known result on stochastic dominance relates expected utility theory and SOSD. This result allows us to translate our problem into an expected utility inequality. A random variable X SOSD the random variable Y if and only if $E_X(u(X)) \geq E_Y(u(Y))$ for all increasing and concave functions u , where the operator $E_X(\cdot)$ represents the expected value with respect to the distribution of X .²⁷

Thus, if the social planner has a utility function that is increasing and concave she will always prefer X to Y if X SOSD Y .²⁸ From the predictive posterior distributions we can get the CDFs and compute the quantity $G_X(a)$ for the treated and for the controls for any plausible range of values for a . From there we can see if the relation (3) holds, if that is the case we can determine wheather the social planner prefers one outcome over the other, all this *without* having to specify any particular functional form for the utility function, other than requiring it to be increasing and concave. Furthermore, there is no need to run any kind of statistical test since we integrated out all the parameter uncertainty.

Figure 6 shows those CDFs by subgroup and Figure 7 shows the graphs for differences between the functions $G_i(\cdot)$ for the treated ($i = 1$) and for the controls ($i = 0$). The CDFs

²⁷For a proof of this result see Green et al. [1995].

²⁸Another characterization of SOSD is that X SOSD Y is equivalent to a mean-preserving spread of the distribution of X in order to obtain the distribution of Y , see Green et al. [1995].

also give a stochastic ranking according to the first order stochastic dominance. However, we could get intersections of the two CDFs for the same subgroup and there is no natural link with utility theory when this happens.²⁹ Thus we look for the groups where the distribution for the treated SOSD the one from the controls. This would look as negative values for the functions in Figure 7 and because of the characterization above this would mean that the expected utility of having that subgroup exposed to the treatment is greater than if the same subgroup is left as a control. This seems to occur in most cases, although in most of them there is indifference at low income levels. These results suggest that groups 3, 10, and 23 would not have been picked to be exposed to the treatment by our social planner. The positive differences appear over the entire interval of observed income. In the cases where the inequality does not hold for the entire interval there is no dominance since the criterion applies when the inequality holds for the entire support of the distribution, which in our case is the range of observed total income. Some of these results contrast with what was obtained by just looking at the CATEs. In that case, only subgroup 3 was not benefited from the program. Subgroups 10 and 23, under the glasses of CATE, have positive mean impacts.

This heterogeneity can be theoretically explained by looking at the different possible scenarios of a person facing either JF or AFDC in an income-leisure space³⁰. A budget line represents the maximum attainable bundles of income and leisure. JF and AFDC however distort this budget line in different ways. AFDC modifies the budget line only at its right lower corner by shifting it upwards by the maximum amount of benefits attainable and by changing the slope of the original budget line up to a certain number of work hours. JF shifts the budget line upwards in a parallel way over a wider interval than AFDC because JF disregards all earnings below the FPL. Depending on where the person finds herself along the AFDC budget line and the shape of her indifference curves, exposure to JF can have

²⁹ X FOSD Y is equivalent to $E_X(u(X)) > E_Y(u(Y))$ for all increasing functions u regardless of their concavity or convexity. So if there are crossings in the CDFs nothing can be said about FOSD, but there could still be SOSD. In other words, FOSD implies SOSD, but SOSD does not necessarily imply FOSD. To summarize, if the CDFs do *not* intersect, for sure there is SOSD, but if they do intersect, there might or might not be SOSD.

³⁰See Bitler et al. [2006]. Leisure is represented in the horizontal axis as work hours in decreasing order, i.e. 0 is to the right of the axis.

different effects on income. A person at a very low income level, exposure to JF can either increase her income or let it unchanged. A person at a top income level would either remain with the same income or reduce her number of working hours, thus lowering her final income in order to maximize her utility.

That explanation works at two different levels. First, two different subgroups contain each homogenous individuals that behave one way or the other as described in the previous paragraph. Second, within a subgroup there is more than one type of individual and the explanation above works for each of them *within* the subgroup. This is an aspect we further discuss in the next section through the use of quantile treatment effects.

5.1 Within-QTEs

Another appealing feature of the method in this paper is that we can recover the quantile treatment effects (QTE) for each subgroup by taking the horizontal difference between the graphs of the two cumulative distribution functions for income. This is shown in Figure 8 with the 90% confidence intervals constructed by bootstrapping³¹. We can think of this as an extension of the heterogeneity analysis of Jobs First in Bitler et al. [2006] except that here we disaggregated that heterogeneity by subgroups. The first thing to remark is that the increasing and then decreasing behavior of the QTEs is not characteristic to all the subgroups as it has been documented in that paper for the entire sample. In some subgroups the QTEs are rather a constant (subgroups 19 and 22) and in a few cases with even negative slope (subgroups 8 and 12). The opt-in effect that Bitler et al. [2006] found at the upper quantiles and then confirmed by Kline and Tartari [2013] using partial identification arguments is not everywhere present across the subgroups. In some of them we find evidence of an opposite effect. We remain agnostic as whether this is driven by behavior not captured before by other methods or whether the QTEs at extreme quantiles are not reliable enough.

These within-QTEs also explain why in some subgroups the CATEs are positive and yet

³¹The method to bootstrap is the same as in Bitler et al. [2006]. We repeat this process 1,000 times. We sample with replacement a sample of the same size as our original sample. We obtain the QTEs. At the end of this bootstrapping process we have 1,000 estimates of QTE for each quantile. At each quantile, sort the QTE estimates from highest to lowest and then choose the 950-th and the 51-st. Those values determine the 90% confidence interval.

the decision making method suggests the opposite. For example, in subgroups 10 and 23 the QTEs are both positive and negative depending on the specific interval of income we are looking at. This means that within these subgroups there is a lot of heterogeneity in the impacts, heterogeneity that is *not* captured by the CATEs, but it *is* captured by the utility theory analysis through the stochastic dominance because it uses the *entire* distribution over the subgroup instead of using one single statistic.

5.2 Robustness Checks

A related paper is the study by Bitler et al. [2010] where they further extend their analysis from their previous work³² to disentangle heterogeneity within- and across-subgroups of the population in a randomized control trial. One of their conclusions is that when only using covariates with information originated at the time of the program, it is not possible to distinguish heterogeneity of impacts across subgroups. However, when using covariates with information pre-program and contemporaneous covariates the opposite occurs. Our results are consistent in the sense that we find substantial heterogeneity across subgroups when using the two types of covariates but not so when we eliminate the use of pre-treatment variables. This is shown in Figure 9 where it is apparent that Jobs First would lead to higher expected utilities than not using the program over almost the entire population.

If we disregard the censoring problem of the data and only use the normal likelihood instead of the Tobit likelihood, the results are qualitatively identical as the ones shown in the tables. The two likelihoods would result in different conclusions if the censoring problem were stronger, for example in the case that we only looked at the earnings variable, since every unemployed person would have an outcome variable equal to 0 and this would create a mass point with a much significant share of the total sample.

As mentioned before, there are other covariates that could be used to evaluate the performance of the program such as a race variable. This would however be the subject of potential conflicts due to basing a decision on the expansion or restriction of the program to a different sample on this type of covariate. We analyze the Jobs First also by substituting

³²Bitler et al. [2006]

the number of kids variable by a race variable that splits the sample into Hispanics or other race, black, and white individuals.³³ In this case four subgroups would not have been picked to be exposed to the treatment by the social planner. This contrasts with the results from the sample CATEs in which only two of those four subgroups had negative impacts. QTEs are relatively constant in four subgroups and only one group exhibits a negatively sloped QTE curve.

Since there are not hyperpriors on the parameters of the prior on the subgroup precisions $\tau_{\mu j}$, we ran a number of different specifications to measure the sensitivity to the hyperparameters α and β . Following our observations in Section 4.1, we ran the entire model with the combinations of parameter values shown in Table 6 and the rest of parameter values as in the main specification. In all of those cases the posterior subgroup means μ_j do not differ from the sample subgroup outcome means by more than 5%. Greater values for α and β would dominate the data in the smallest subgroups. None of our results on SOSD changes for any of those alternative specifications except for subgroups 7 and 10 when the ratio α/β is relatively large. This might be the result of an exacerbated ratio in the variance for the conditional posterior distribution for $\tau_{\mu j}$ since the denominator, when $\epsilon \rightarrow 0$, becomes considerably smaller with respect to the numerator.

6 Conclusions

Heterogeneity of treatment effects is a relevant issue when extending or reducing the target population of a welfare program. On one side there has been a literature exploring non-parametric methods to measure this heterogeneity. On the other side a rather small literature has focused on mapping outcome distributions into expected utility theory criteria to make decisions. This paper combines these two literatures to develop a method that finds predictive outcome distributions conditional on covariates for treated and controls and maps those into second order stochastic dominance criteria. This gives a ranking of treatment versus control for each subgroup of the sample without having to specify the utility functional form other

³³These results are available upon request.

than minimal requirements. This can be used to assess whether to extend or restrict the program to a larger or smaller sample to maximize welfare.

We use a Bayesian approach since the size of the subsamples can be very small and classical approaches tend to not be reliable in these cases. Another issue is that the effect of the program in each subgroup is potentially correlated with other subgroups' effects since there might be covariates that make them similar and we did not capture that when dividing into the subsamples. Our hierarchical Bayesian model allows for correlation across subgroups.

We find that one eighth of the number of subgroups in which we divided our sample does not maximize expected utility when exposed to the treatment. Only one of those such subgroups shows negative impacts using non-parametric methods. This is because our method sheds light on the benefits from being assigned to the program not just from a point-estimation perspective, but by using the entire distribution of predictive outcomes in order to make a decision. As a by-product we can also calculate quantile treatment effects by subgroup showing that in most of them upper income levels are more benefited from the program. This within-variation in part explains the discrepancy between the non-parametric methods and our approach. Even though we have information on race, we do not believe it could successfully be used as a profiling covariate when extending or restricting a program to a different sample. For that reason we use other covariates to form the subgroups that have been used in the literature before but show the results as well as if the race covariate had been used. We let for future research the endogeneity of the number of subgroups and how to split the sample. One possible solution for this is to use a latent class model to determine the probability of each individual of belonging to each subgroup given an exogenous number of subgroups.

Finally, learning about the differences in impacts across subsamples in a decision making framework can guide policymakers towards welfare improving modifications to the design of welfare programs.

Appendix

Gibbs sampler

To get the posterior distribution of one of the subgroup means, we condition on all the other variables and the parameters of the model, where μ_{-j} is the collection of the μ_k with $k \neq j$, then

$$\begin{aligned}
p(\mu_j|A) &\propto \left(\prod_{i=1}^{n_j} \exp\left(-\frac{1}{2}\tau_{\mu j}(y_{ij} - \mu_j)^2\right)\right) \times \exp\left(-\frac{1}{2}h_1(\mu_j - \mu)^2\right) \\
&= \exp\left(-\frac{1}{2}\tau_{\mu j} \sum_{i=1}^{n_j} (y_{ij} - \mu_j)^2 - \frac{1}{2}h_1(\mu_j - \mu)^2\right) \\
&= \exp\left(-\frac{1}{2}\tau_{\mu j} \left(\sum_{i=1}^{n_j} y_{ij}^2 + n_j\mu_j^2 - 2\mu_j \sum_{i=1}^{n_j} y_{ij}\right) - \frac{1}{2}h_1(\mu_j^2 + \mu^2 - 2\mu_j\mu)\right) \\
&\propto \exp\left(\left(-\frac{1}{2}\tau_{\mu j}n_j - \frac{1}{2}h_1\right)\mu_j^2 - \frac{1}{2}\left(-2\tau_{\mu j} \sum_{i=1}^{n_j} y_{ij} - 2h_1\mu\right)\mu_j\right) \\
&\propto \exp\left(-\frac{1}{2}(\tau_{\mu j}n_j + h_1)\left(\mu_j - \frac{\tau_{\mu j} \sum_{i=1}^{n_j} y_{ij} + h_1\mu}{\tau_{\mu j}n_j + h_1}\right)^2\right), \text{ for } j = 1, \dots, J
\end{aligned}$$

so that each mean subgroup has a normal posterior distribution,

$$\mu_j|A \sim N\left(\frac{\tau_{\mu j} \sum_{i=1}^{n_j} y_{ij} + h_1\mu}{\tau_{\mu j}n_j + h_1}, \tau_{\mu j}n_j + h_1\right), \text{ for } j = 1, \dots, J$$

or equivalently,

$$\mu_j|A \sim N\left(\frac{\sum_{i=1}^{n_j} y_{ij} + h_1\mu/\tau_{\mu j}}{n_j + h_1/\tau_{\mu j}}, \tau_{\mu j}n_j + h_1\right), \text{ for } j = 1, \dots, J.$$

We can see that the posterior mean is a weighted average of the observations within the group and μ . The precision is a weighted sum of the precision of the income for subgroup j and the precision of μ_j .

Now, let's calculate the posterior distribution of μ . From the joint distribution, condi-

tioning on the other variables we have that

$$\begin{aligned}
p(\mu|A) &\propto \left(\prod_j \exp \left(-\frac{1}{2} h_1 (\mu_j - \mu)^2 \right) \right) \times \exp \left(-\frac{1}{2} h_2 \mu^2 \right) \\
&= \exp \left(-\frac{1}{2} h_1 \left(\sum_{j=1}^J \mu_j^2 - 2\mu \sum_{j=1}^J \mu_j + J\mu^2 \right) - \frac{1}{2} h_2 \mu^2 \right) \\
&\propto \exp \left(-\frac{1}{2} h_1 \left(-2\mu \sum_{j=1}^J \mu_j + J\mu^2 \right) - \frac{1}{2} h_2 \mu^2 \right) \\
&= \exp \left(-\frac{1}{2} \left((Jh_1 + h_2) \mu^2 - 2\mu h_1 \sum_{j=1}^J \mu_j \right) \right) \\
&\propto \exp \left(-\frac{1}{2} (Jh_1 + h_2) \left(\mu - \frac{h_1 \sum_{j=1}^J \mu_j}{Jh_1 + h_2} \right)^2 \right)
\end{aligned}$$

so the hyperparameters are

$$\mu|A \sim N \left(\frac{h_1 \sum_{j=1}^J \mu_j}{Jh_1 + h_2}, Jh_1 + h_2 \right),$$

or equivalently,

$$\mu|A \sim N \left(\frac{\sum_{j=1}^J \mu_j}{J + h_2/h_1}, Jh_1 + h_2 \right),$$

the posterior mean is a weighted average of the subgroup means and the posterior precision is a weighted sum of the precisions of the subgroup means and the precision of μ . Also note that these hyperparameters do not depend explicitly on the value of the observations.

The posterior distribution of $\tau_{\mu j}$.

$$\begin{aligned}
p(\tau_{\mu j}|A) &\propto \tau_{\mu j}^{n_j/2} \exp \left(-\frac{1}{2} \tau_{\mu j} \sum_{i \in j} (y_{ij} - \mu_j)^2 \right) \tau_{\mu j}^{\alpha-1} \exp(-\beta \tau_{\mu j}) \\
&= \tau_{\mu j}^{\alpha+n_j/2-1} \exp \left(-\left(\beta + \frac{1}{2} \sum_{i \in j} (y_{ij} - \mu_j)^2 \right) \tau_{\mu j} \right), \text{ for } j = 1, \dots, J.
\end{aligned}$$

This expression is the kernel of a gamma distribution, the hyperparameters are

$$\tau_{\mu}|A \sim G \left(\alpha + n_j/2, \beta + \frac{1}{2} \sum_{i \in j} (y_{ij} - \mu_j)^2 \right).$$

Observe that these hyperparameters do not depend on h_1 or h_2 in the conditional posterior distribution, but in the number of total observations and the overall sum of the squares of the deviations from the subgroup means.

Remember that we gave a prior on h_1 , the precision of the group means μ_j ,

$$p(h_1) \propto h_1^{\gamma-1} \exp(-\gamma h_1).$$

If A represents all the other parameters, the full conditional distribution for h_1 is

$$\begin{aligned} p(h_1|A) &\propto h_1^{J/2} \exp\left(-\frac{1}{2}h_1 \sum_{j=1}^J (\mu_j - \mu)^2\right) \times h_1^{\gamma-1} \exp(-\gamma h_1) \\ &= h_1^{\gamma+J/2-1} \exp\left(-h_1 \left(\gamma + \frac{1}{2} \sum_{j=1}^J (\mu_j - \mu)^2\right)\right) \end{aligned}$$

so $h_1|A \sim G(\gamma+J/2, \gamma+\frac{1}{2} \sum_{j=1}^J (\mu_j - \mu)^2)$. In the actual estimation, $\gamma = 0.1$. For h_2 we assume exactly the same prior as for h_1 but its posterior distribution is $h_2|A \sim G(\gamma + 1/2, \gamma + \frac{1}{2}\mu^2)$ and $\gamma = 0.1$. Finally, ν was given the prior $\nu \sim N(0, h_3)$. Its posterior distribution is $\nu|A \sim N(\frac{\mu}{1+h_3/h_2}, h_3 + h_2)$. Note that if h_2 did not have a posterior distribution then h_3/h_2 would be a constant. The initial parameters for the model are specified in Table 5.

Tobit model

In the case of a mass point at $y = a$ the strategy to be adopted is a Tobit model (Dehejia [2005]). To simplify the notation we will write everything as if $a = 0$. With this we transform the likelihood to be the one of a censored normal distribution. With this new likelihood there is a mass point on 0 and then a truncated normal distribution for the rest of the observations. This likelihood has the form

$$p(y_{ij}|A) = \prod_{i:y_{ij}=0} (1 - \Phi(\mu_j \tau_\mu^{1/2})) \times \prod_{i:y_{ij}>0} \tau_\mu^{1/2} \exp\left(-\frac{1}{2}\tau_\mu (y_{ij} - \mu_j)^2\right).$$

Where A represents all the parameters of the model and Φ is the normal cumulative distribution function. The priors are exactly the same as before. Following the idea of Chib [1992] for Bayesian estimation of a model with censored observations, we can transform the likelihood

into a product of normals as before. To do so, let's assume that there exist (hypothetical) negative observations of y_{ij} when we observe $y_{ij} = 0$. We can do that by simulating negative observations from truncated normal distributions with parameters μ_j and τ_μ . Now the mass point over 0 has been substituted by a collection of hypothetical negative observations that contribute to the likelihood as normal densities, leading to the same expressions at the beginning of this section.

Assume that the collection of observations for Y is $\{\tilde{y}_{ij}\}$. But we observe the censored sample³⁴

$$y_{ij} = \begin{cases} \tilde{y}_{ij} & \text{if } \tilde{y}_{ij} > 0, \\ 0 & \text{if } \tilde{y}_{ij} \leq 0. \end{cases}$$

There is a known, set valued function C_{ij} so that

$$C_{ij} = \begin{cases} \tilde{y}_{ij} & \text{if } \tilde{y}_{ij} > 0, \\ (-\infty, 0] & \text{if } \tilde{y}_{ij} \leq 0. \end{cases}$$

Then the joint distribution of observables and unobservables is

$$\begin{aligned} p(y_{ij}, \mu, \{\mu_j\}_{j=1,\dots,J}, \{\tau_{\mu j}\}_{j=1,\dots,J} | A) &= p(y_{ij} | \mu, \mu_j, \tau_{\mu j}, A) p(C | y_{ij}, A) \times \\ &\times p(\mu | A) p(\{\mu_j\}_{j=1,\dots,J} | A) p(\{\tau_{\mu j}\}_{j=1,\dots,J} | A) \end{aligned}$$

where

$$p(C | y_{ij}, A) = \prod_{i,j} p(C_{ij} | y_{ij}, A) = \prod_{i,j} 1_{C_{ij}}(y_{ij}).$$

So that the joint distribution is proportional to

$$\begin{aligned} &\prod_{i,j} 1_{C_{ij}}(y_{ij}) \times \left(\prod_{i,j} \tau_{\mu j}^{1/2} \exp \left(-\frac{1}{2} \tau_{\mu j} (y_{ij} - \mu_j)^2 \right) \right) \times \\ &\left(\prod_{j=1}^J h_1^{1/2} \exp \left(-\frac{1}{2} h_1 (\mu_j - \mu)^2 \right) \right) \times \\ &h_2^{1/2} \exp \left(-\frac{1}{2} h_2 \mu^2 \right) \times \\ &\prod_{j=1}^J \tau_{\mu j}^{\alpha-1} \exp(-\beta \tau_{\mu j}). \end{aligned}$$

³⁴As in Geweke [2005]

This expression is almost identical to the one presented at the beginning of this section. The censoring data is not a problem in the model because the posterior distributions will have the same kernels as before. But now the posterior kernels include those y_{ij} 's that are negative too. Then the posterior distributions will be slightly different.

The Gibbs sampler for this likelihood is like the one from the previous section except that now there is a data augmentation step. The Gibbs sampler from the previous section is now nested in the data augmentation step. In the first iteration, negative values are drawn from a truncated normal distribution to replace the observations of zero income. Then the Gibbs sampler from the previous section is carried out exactly as in the normal model. In order to get predictive distributions we need to slightly modify the algorithm from the normal model:

1. Get a random draw from the posterior distributions $p(\mu_j|A)$ and $p(\tau_{\mu_j}|A)$.
2. Get a random draw from

$$p(y_{ij}|A) = \prod_{i:y_{ij}=0} (1 - \Phi(\mu_j \tau_{\mu_j}^{1/2})) \times \prod_{i:y_{ij}>0} \tau_{\mu_j}^{1/2} \exp\left(-\frac{1}{2}\tau_{\mu_j}(y_{ij} - \mu_j)^2\right).$$

The last step can be done for treated and controls and for each $j = 1, \dots, J$ this way:

1. get a random number u from a uniform density on $[0, 1]$.
2. if $u < 1 - \Phi(\mu_j \tau_{\mu_j}^{1/2})$ then $y_{ij} = 0$. otherwise, get a random draw from a truncated normal on $[0, \infty)$ with the appropriate parameters.

Tables and Figures

Table 1: Subgroups definitions

group	age > 30	Prev. employed	0 or 1 kids	2 kids	3+ kids	high school degree
1			X			
2				X		
3					X	
4	X		X			
5	X			X		
6	X				X	
7		X	X			
8		X		X		
9		X			X	
10	X	X	X			
11	X	X		X		
12	X	X			X	
13			X			X
14				X		X
15					X	X
16	X		X			X
17	X			X		X
18	X				X	X
19		X	X			X
20		X		X		X
21		X			X	X
22	X	X	X			X
23	X	X		X		X
24	X	X			X	X

Note: Subgroups are exogenously defined by covariates used in similar studies.

Table 2: Subgroup treatment effects by OLS

Covariate	Coefficient	Std Error	t-stat	p-value
e	758.932	324.227*	2.34	0.02
g-j1	-1,033.282	306.850**	-3.37	0.00
g-j2	-562.983	314.462	-1.79	0.07
g-j3	228.297	332.997	0.69	0.49
g-j4	-1,162.854	301.553**	-3.86	0.00
g-j5	-636.512	312.033*	-2.04	0.04
g-j6	-135.569	304.579	-0.45	0.66
g-j7	-299.680	286.975	-1.04	0.30
g-j8	249.002	313.456	0.79	0.43
g-j9	607.030	342.940	1.77	0.08
g-j10	170.884	303.409	0.56	0.57
g-j11	85.352	312.033	0.27	0.78
g-j12	399.921	316.629	1.26	0.21
g-j13	-867.300	316.629**	-2.74	0.01
g-j14	-515.989	335.210	-1.54	0.12
g-j16	-951.523	304.279**	-3.13	0.00
g-j17	-602.223	321.009	-1.88	0.06
g-j18	141.217	319.672	0.44	0.66
g-j19	33.578	283.170	0.12	0.91
g-j20	365.124	314.462	1.16	0.25
g-j21	635.265	347.569	1.83	0.07
g-j22	-269.681	301.553	-0.89	0.37
g-j23	319.663	301.062	1.06	0.29
g-j24	618.572	311.580*	1.99	0.05
e-g-j1	-537.150	374.053	-1.44	0.15
e-g-j2	-546.896	390.099	-1.40	0.16
e-g-j3	-817.683	411.756*	-1.99	0.05
e-g-j4	-524.877	374.519	-1.40	0.16
e-g-j5	-286.250	386.018	-0.74	0.46
e-g-j6	-323.625	376.085	-0.86	0.39
e-g-j7	-626.252	349.531	-1.79	0.07
e-g-j8	-452.906	397.668	-1.14	0.25
e-g-j9	-523.388	432.720	-1.21	0.23
e-g-j10	-714.583	375.569	-1.90	0.06
e-g-j11	-261.970	386.593	-0.68	0.50
e-g-j12	65.474	395.907	0.17	0.87
e-g-j13	-367.914	389.743	-0.94	0.35
e-g-j14	-337.058	445.510	-0.76	0.45
e-g-j15	-771.915	509.051	-1.52	0.13
e-g-j16	-345.271	384.074	-0.90	0.37
e-g-j17	-392.305	407.773	-0.96	0.34
e-g-j18	-266.634	392.499	-0.68	0.50
e-g-j19	-629.959	345.113	-1.83	0.07
e-g-j20	-134.819	393.759	-0.34	0.73
e-g-j22	-150.651	369.871	-0.41	0.68
e-g-j23	-666.457	378.408	-1.76	0.08
e-g-j24	-11.050	391.052	-0.03	0.98
constant	2,621.703	271.891**	9.64	0.00
F	16.18			
R ² -adj	0.14			
N	4,463			

Note: A simple OLS estimation of CATEs by projecting the outcome variable onto the treatment variable e , subgroup dummies $g-j$, and interactions of subgroup dummies and the treatment variable $e-g-j$. Statistical significance is rarely found. * $p < 0.05$; ** $p < 0.01$.

Table 3: Sample means

group	n	T = 1		n	T = 0		sample CATE		t stat.
		sample mean			sample mean				
1	129	1,852	(1,072)	91	1,658	(980)	194	(142)	1.4
2	85	2,324	(1,103)	76	2,086	(857)	238	(157)	1.5
3	69	2,832	(1,322)	51	2,906	(1,120)	-74	(229)	-0.3
4	104	1,725	(1,054)	108	1,526	(900)	199	(134)	1.5
5	89	2,596	(1,235)	80	2,035	(1,019)	561	(175)	3.2
6	109	2,948	(1,493)	99	2,561	(1,270)	387	(193)	2.0
7	222	2,466	(1,212)	227	2,332	(1,127)	134	(110)	1.2
8	66	3,225	(1,170)	79	2,871	(1,486)	354	(225)	1.6
9	48	3,609	(1,360)	44	3,229	(1,193)	380	(268)	1.4
10	108	2,837	(2,254)	105	2,819	(2,119)	18	(300)	0.1
11	92	3,204	(1,808)	82	2,707	(1,373)	497	(246)	2.0
12	76	3,846	(1,928)	72	3,064	(2,002)	782	(323)	2.4
13	91	2,216	(1,240)	68	1,883	(854)	333	(175)	1.9
14	35	2,528	(1,317)	49	2,149	(1,065)	379	(260)	1.5
15	23	2,722	(1,143)	26	2,622	(1,016)	100	(308)	0.3
16	81	2,084	(1,261)	103	1,670	(956)	414	(163)	2.5
17	60	2,386	(1,236)	65	2,051	(971)	336	(198)	1.7
18	92	3,291	(1,469)	68	2,763	(1,205)	528	(218)	2.4
19	249	2,784	(1,371)	306	2,664	(1,326)	120	(115)	1.0
20	77	3,611	(1,459)	77	2,987	(1,026)	624	(203)	3.1
21	33	4,016	(1,354)	41	3,257	(1,127)	759	(288)	2.6
22	130	2,983	(1,477)	112	2,373	(1,488)	610	(191)	3.2
23	90	3,034	(1,901)	115	2,941	(1,360)	92	(228)	0.4
24	78	3,988	(1,762)	83	3,240	(1,566)	748	(262)	2.9

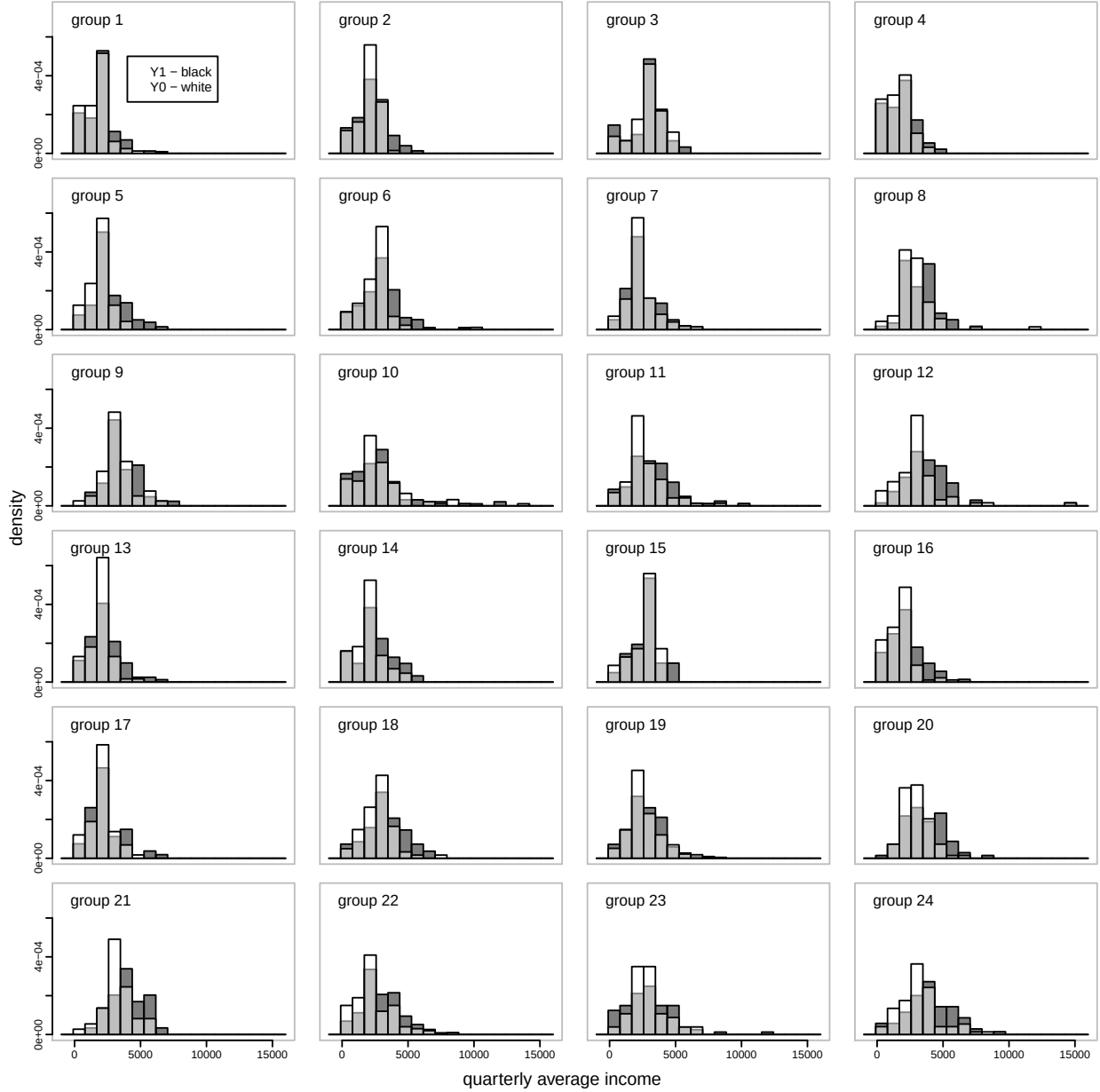
Note: Sample standard errors in parentheses. See the main text for details on the sample CATE standard error calculation. Total sample size for $T = 1$ is 2,227 and for $T = 0$ is 2,236.

Table 4: Posterior CATE means by subgroup

group	n T = 1	n T = 0	sample CATE		posterior mean CATE	
1	129	91	194	(142)	183	(137)
2	85	76	238	(157)	243	(155)
3	69	51	-74	(229)	-43	(218)
4	104	108	199	(134)	203	(130)
5	89	80	561	(175)	557	(170)
6	109	99	387	(193)	385	(185)
7	222	227	134	(110)	143	(109)
8	66	79	354	(225)	373	(210)
9	48	44	380	(268)	396	(265)
10	108	105	18	(300)	67	(288)
11	92	82	497	(246)	490	(240)
12	76	72	782	(323)	769	(291)
13	91	68	333	(175)	338	(169)
14	35	49	379	(260)	399	(256)
15	23	26	100	(308)	135	(299)
16	81	103	414	(163)	429	(170)
17	60	65	336	(198)	344	(206)
18	92	68	528	(218)	527	(209)
19	249	306	120	(115)	126	(111)
20	77	77	624	(203)	599	(195)
21	33	41	759	(288)	677	(285)
22	130	112	610	(191)	584	(192)
23	90	115	92	(228)	98	(222)
24	78	83	748	(262)	719	(260)

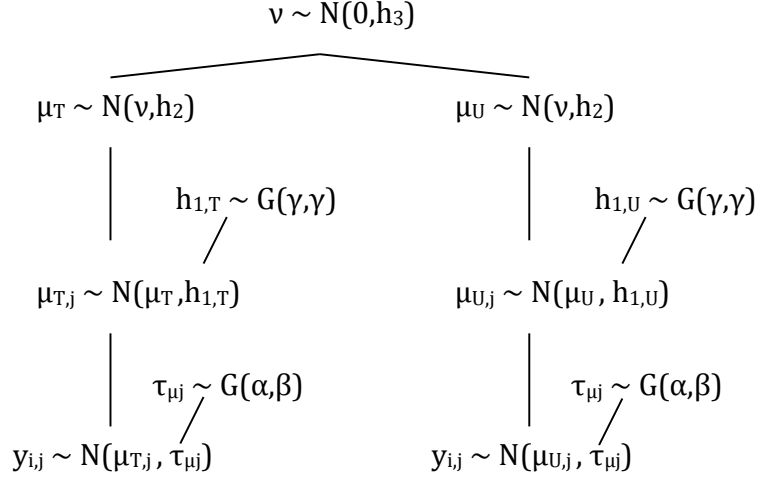
Note: Contrasts between the sample CATE estimates and the CATEs obtained by taking the difference $\mu_j^{treated} - \mu_j^{control}$ using the random draws from the Gibbs sampler. See main text for details. Tobit likelihood specification. Sample standard errors in parentheses.

Figure 1: Income data histograms by subgroup for treated and controls



Note: Each panel shows for each subgroup and for both treated and controls the empirical distribution of the outcome variable, defined as earnings plus transfers. There is a mass point on zero due to the 1.15% of the sample that did not receive any income nor transfers.

Figure 2: Diagram of hierarchical Bayesian model

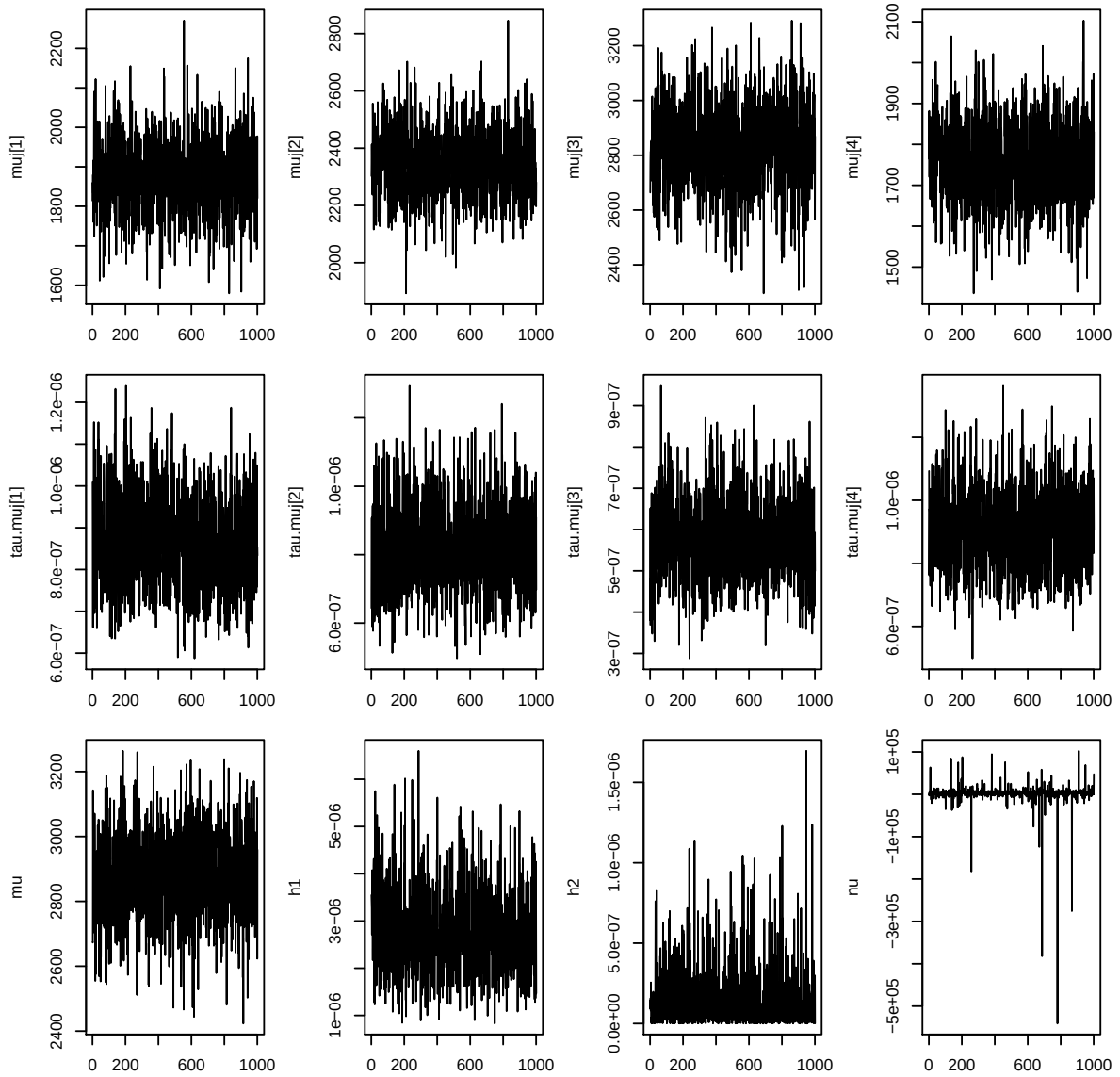


Note: The hyperparameters h_2 and ν also have a prior. There are actually J branches one for each subgroup at the level of the $\mu_{\cdot,j} \sim N(\mu_{\cdot}, h_{1,\cdot})$ distribution but for simplicity we only illustrate one of them. The same applies to the bottom level but with as many branches as data points in each subgroup. See main text for details.

Table 5: Initial parameter values

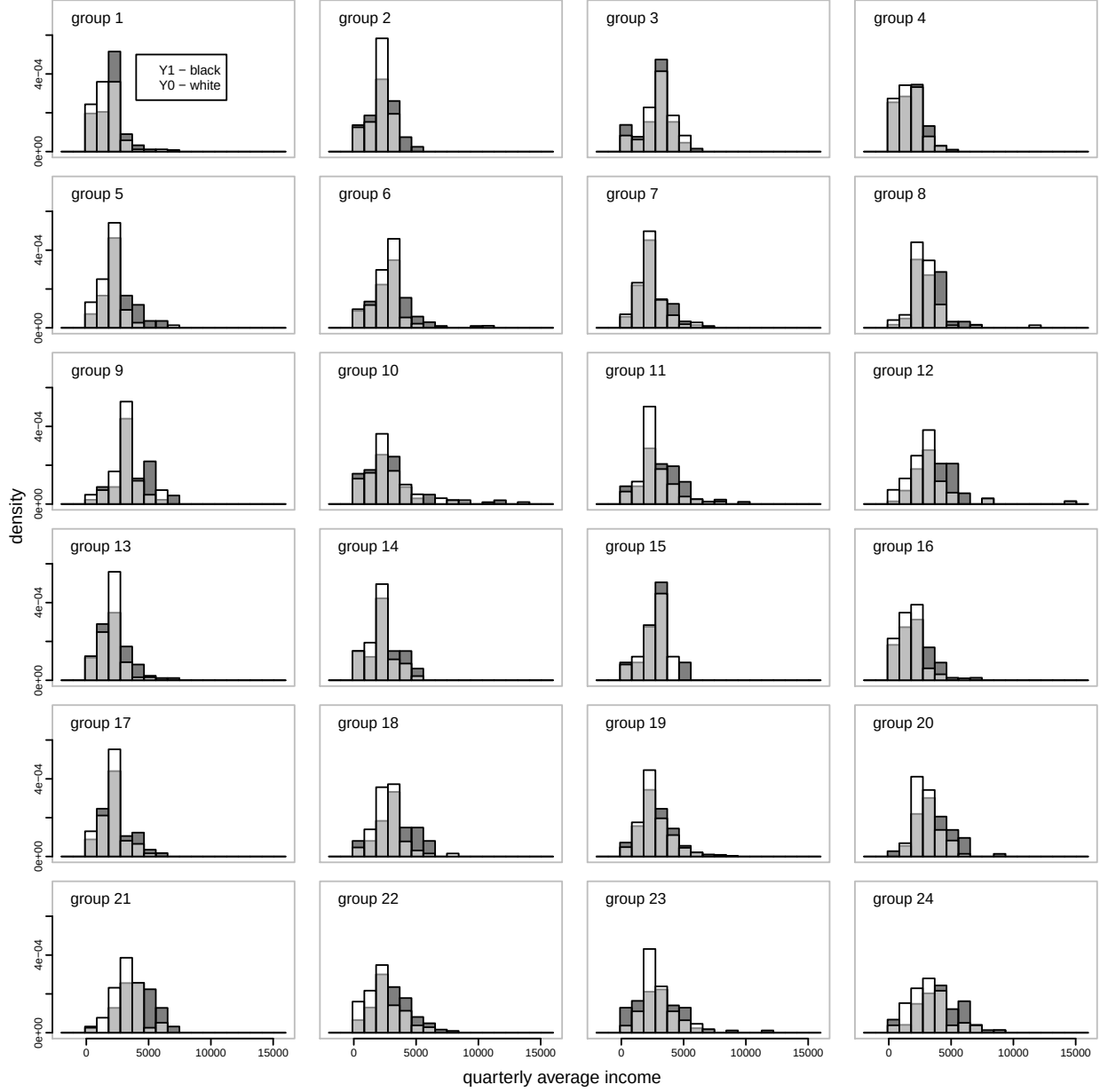
Parameter	Value
α	0.1
β	0.1
h_3	$1E-12$
γ	0.1

Figure 3: Times series of random draws from the Gibbs sampler



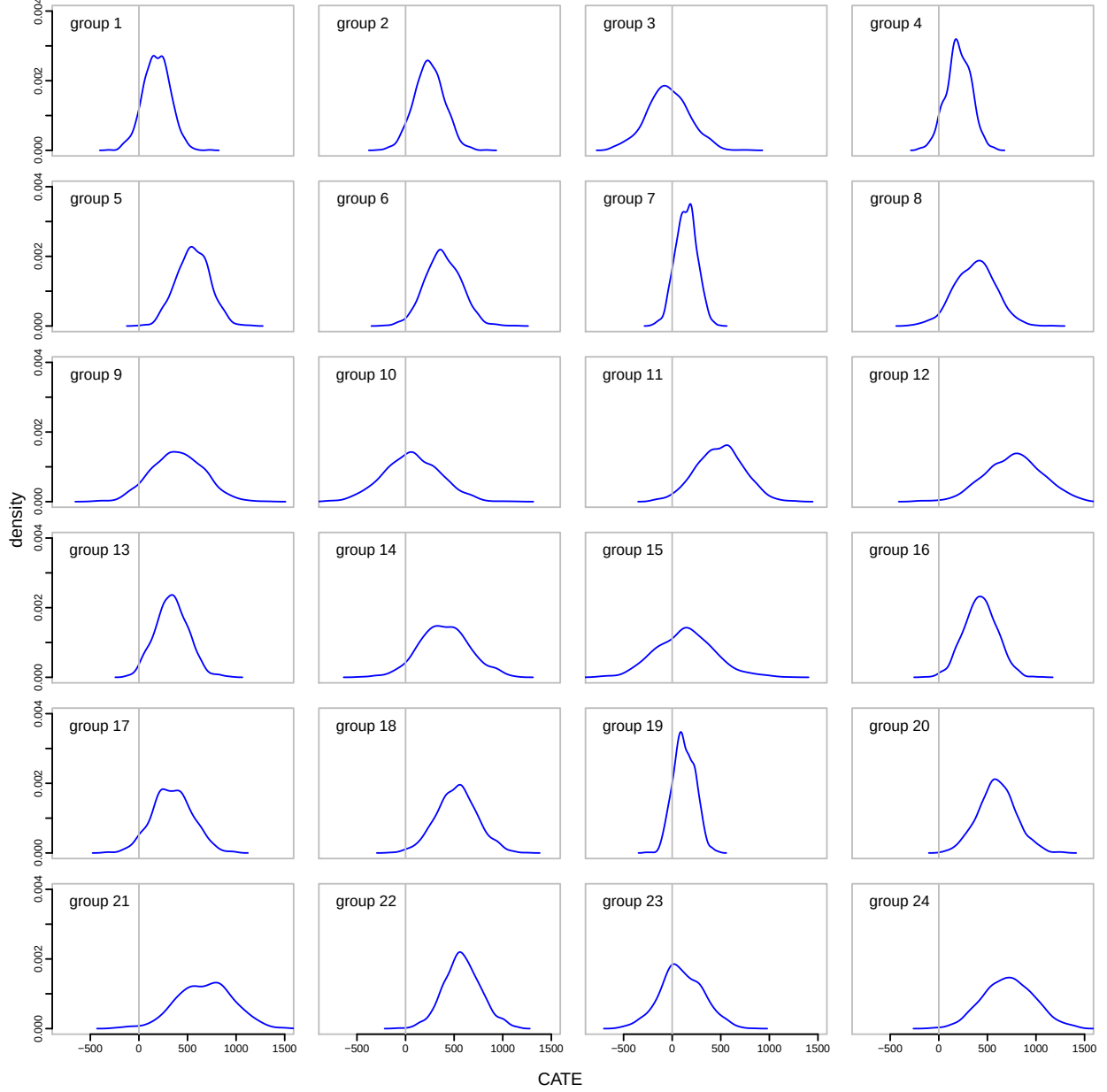
Note: Each graph shows the time series of random draws that form the posterior distributions. These are only the *last* 1,000 random draws from the Gibbs sampler for selected parameters from the treated subsample.

Figure 4: Income histograms of predictive income distributions



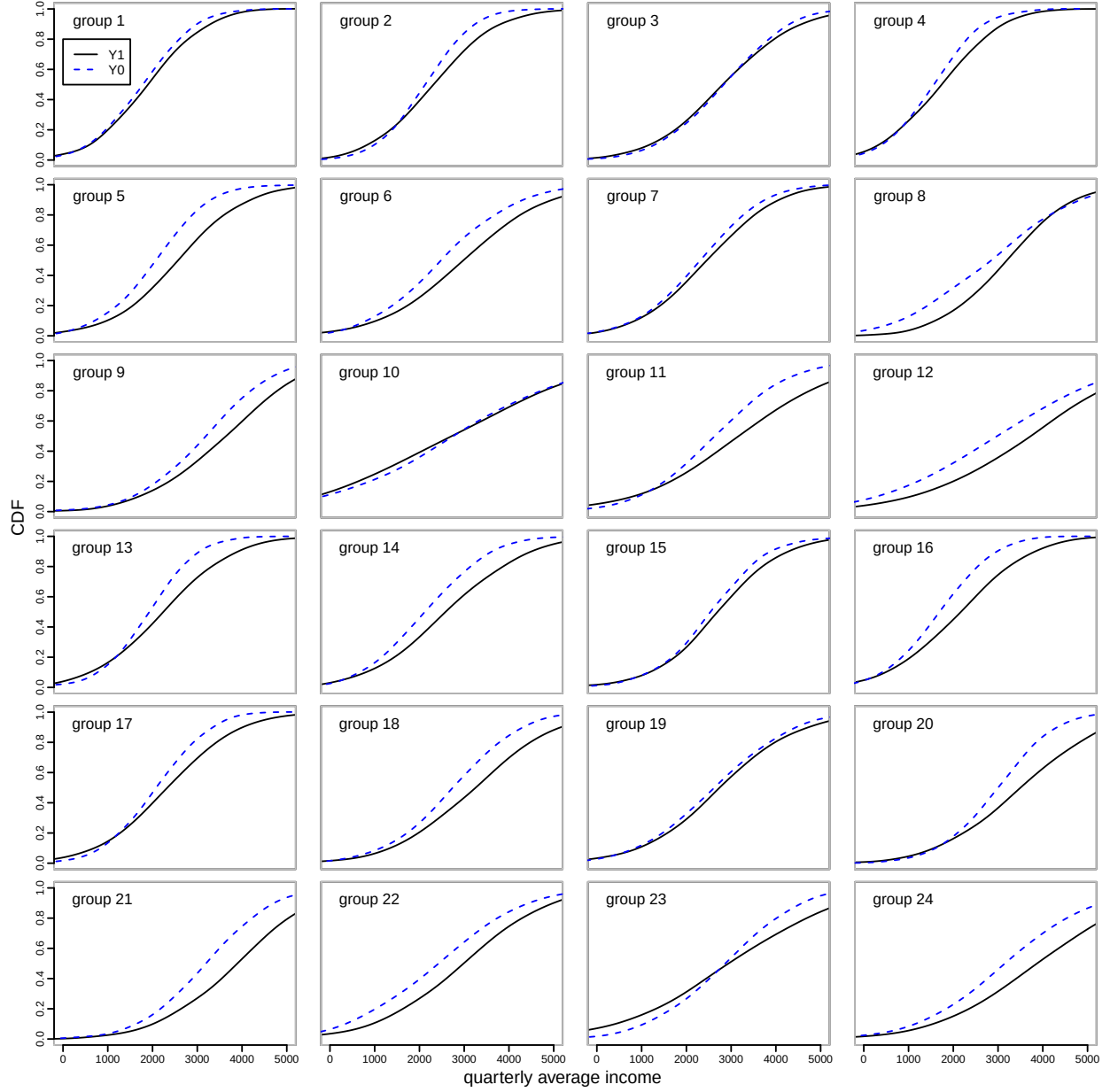
Note: For each subgroup and for both treated and controls we calculate predictive income distributions. These histograms therefore reflect the distributions *after* the data augmentation step. See main text for details.

Figure 5: Kernel estimates for the CATEs posterior densities by subgroup



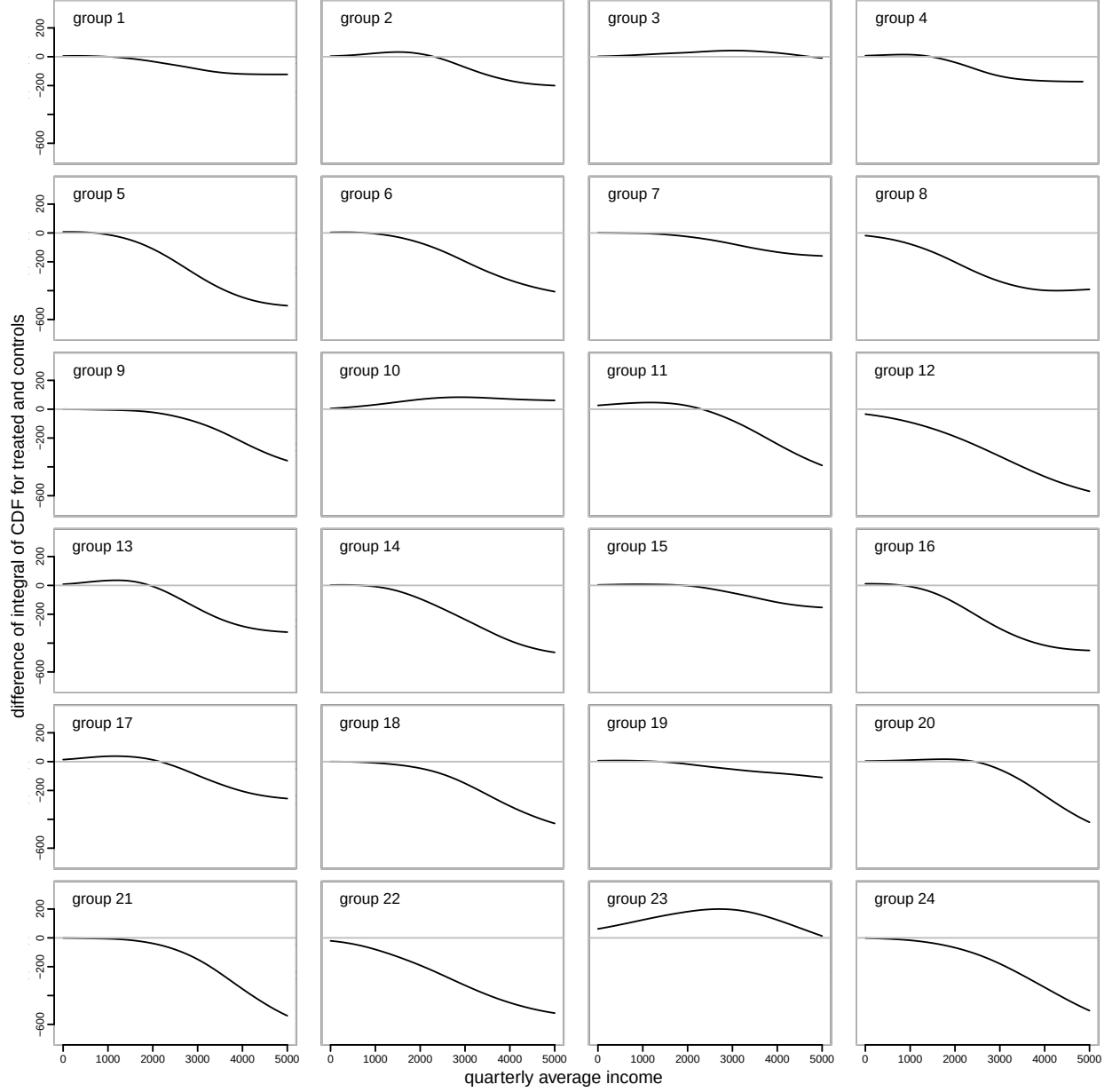
Note: Each graph represents the kernel density estimate for the distribution of the CATE for that subgroup obtained by taking the difference $\mu_j^{treated} - \mu_j^{control}$ using the random draws from the Gibbs sampler. See main text for details.

Figure 6: Cumulative distribution functions of predictive income



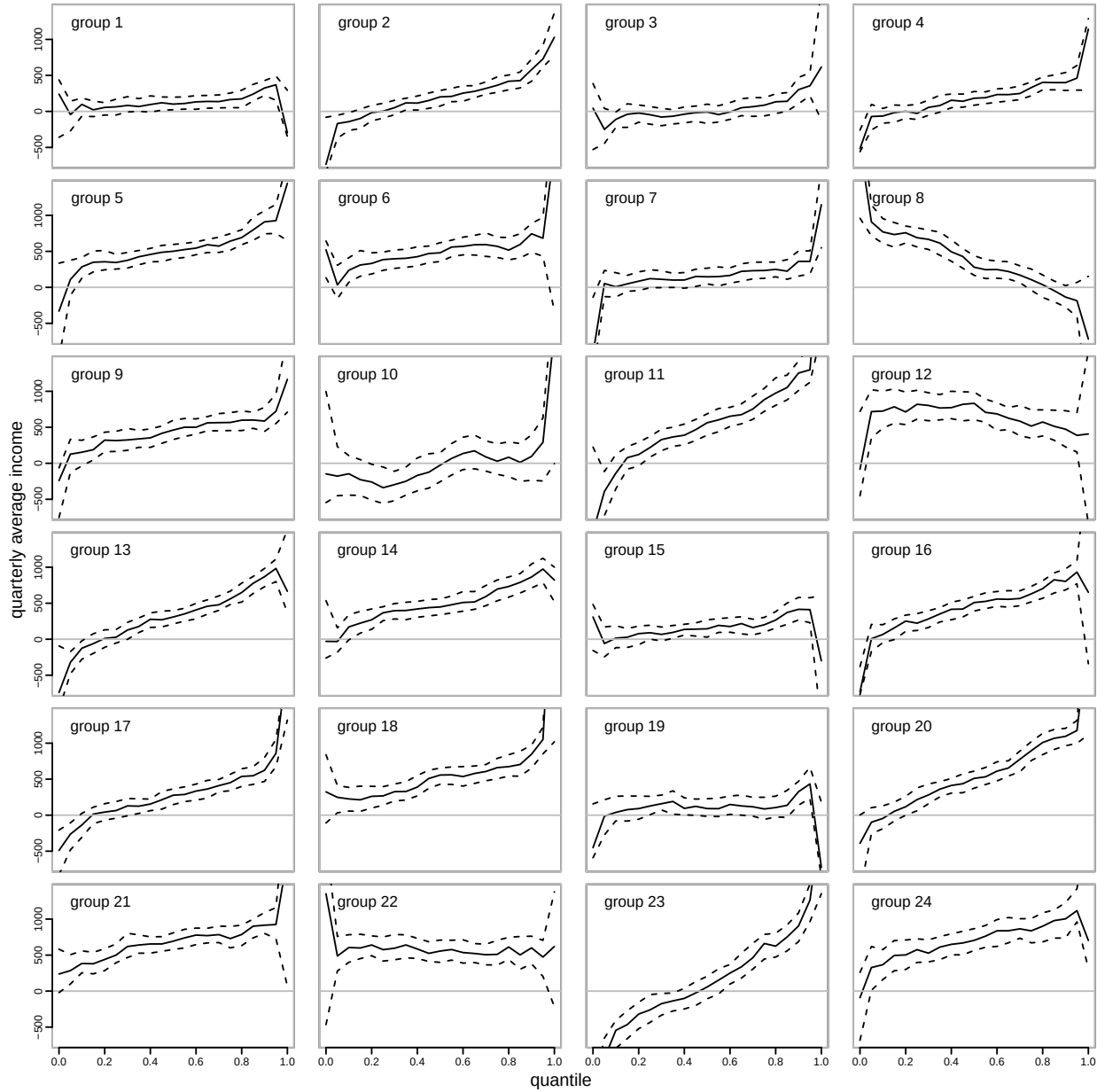
Note: Each graph represents the cumulative distribution functions of income for treated and controls over an interval of quarterly average income that covers the data. First order stochastic dominance of being treated is evident in most subgroups. The horizontal difference of the two lines for a given subgroup and for a given number in the vertical axis is the QTE.

Figure 7: Second order stochastic dominance by subgroup



Note: Each graph represents the difference between the graphs of $G_X(a) = \int^a F_X(t)dt$ where F_X is the predictive distribution's CDF for income (treatment minus controls). If the resulting difference is negative then there is SOSD of being treated with respect to being assigned to the control group.

Figure 8: Quantile treatment effects by group



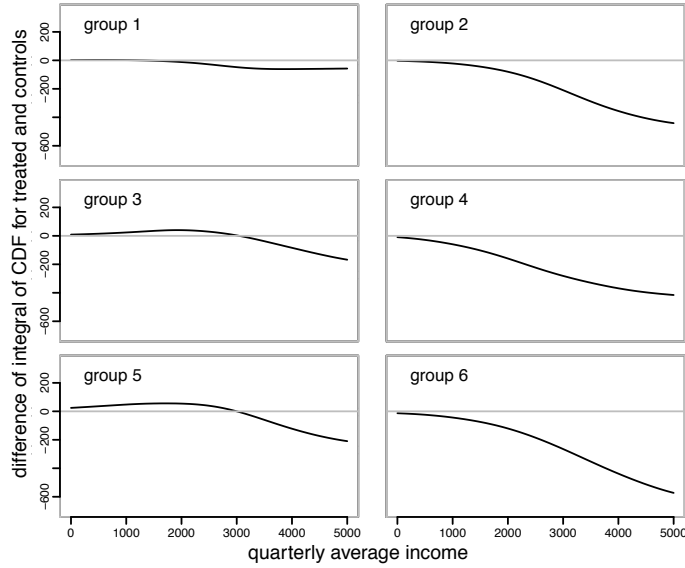
Note: Continuous lines represent the QTEs over the income distribution. Bootstrapped 90% confidence intervals in dashed lines. See text for details on the bootstrapping process.

Table 6: Initial parameter values

α	β	Subgroups*
0.1	0.1	3, 10, 23
0.2	0.2	3, 7, 10, 23
1	0.2	3, 23
10	2	3, 23
2	2	3, 23
5	2	3, 7, 10, 23

Note: These subgroups exhibit SOSD from being exposed to the control.

Figure 9: Second order stochastic dominance by subgroup without pre-treatment information



Note: Each graph represents the difference between the graphs of $G_X(a) = \int^a F_X(t)dt$ where F_X is the predictive distribution's CDF for income (treatment minus controls) without using pre-treatment variables. Subgroup definitions: 1: age < 30, pregnant or 1 kid; 2: age < 30, 1 kid; 3: age < 30, 3 or more kids; 4: age $\geq$ 30, pregnant or 1 kid; 5: age $\geq$ 30, 1 kid; 6: age $\geq$ 30, 3 or more kids.

Table 7: Posterior means

subgroup	μ_T	s.d.	τ_T	s.d.	μ_U	s.d.	τ_U	s.d.
1	1,874	95	8.69E-07	1.10E-07	1,691	100	1.04E-06	1.53E-07
2	2,345	118	8.21E-07	1.27E-07	2,102	96	1.36E-06	2.08E-07
3	2,833	153	5.73E-07	9.63E-08	2,876	150	7.92E-07	1.64E-07
4	1,759	98	9.05E-07	1.29E-07	1,556	87	1.23E-06	1.69E-07
5	2,609	126	6.57E-07	9.82E-08	2,053	114	9.68E-07	1.50E-07
6	2,942	143	4.49E-07	5.84E-08	2,557	118	6.22E-07	8.99E-08
7	2,474	80	6.82E-07	6.51E-08	2,331	75	7.90E-07	7.29E-08
8	3,202	138	7.35E-07	1.23E-07	2,830	165	4.57E-07	7.31E-08
9	3,540	190	5.39E-07	1.16E-07	3,144	180	7.02E-07	1.50E-07
10	2,841	211	1.97E-07	2.73E-08	2,774	191	2.24E-07	3.14E-08
11	3,176	181	3.10E-07	4.53E-08	2,686	150	5.33E-07	8.35E-08
12	3,733	218	2.70E-07	4.58E-08	2,955	209	2.51E-07	4.28E-08
13	2,250	128	6.47E-07	9.76E-08	1,911	106	1.37E-06	2.38E-07
14	2,575	220	5.81E-07	1.38E-07	2,176	148	8.89E-07	1.80E-07
15	2,740	238	7.73E-07	2.24E-07	2,607	203	9.70E-07	2.71E-07
16	2,123	144	6.31E-07	9.84E-08	1,694	92	1.09E-06	1.59E-07
17	2,422	161	6.54E-07	1.20E-07	2,078	122	1.06E-06	1.84E-07
18	3,265	154	4.60E-07	6.73E-08	2,738	143	6.88E-07	1.19E-07
19	2,788	85	5.33E-07	5.04E-08	2,662	75	5.70E-07	4.66E-08
20	3,560	161	4.71E-07	7.58E-08	2,961	113	9.48E-07	1.51E-07
21	3,855	230	5.41E-07	1.38E-07	3,172	173	7.86E-07	1.74E-07
22	2,975	132	4.59E-07	5.41E-08	2,391	133	4.53E-07	6.01E-08
23	3,015	182	2.77E-07	4.04E-08	2,918	125	5.43E-07	7.38E-08
24	3,885	206	3.22E-07	5.18E-08	3,163	164	4.04E-07	6.50E-08
	$T = 1$	μ	s.d.	h_1	s.d.	h_2	s.d.	
		2,870	134	2.71E-06	8.71E-07	1.52E-07	1.93E-07	
	$T = 0$	μ	s.d.	h_1	s.d.	h_2	s.d.	
		2,500	115	3.92E-06	1.19E-06	2.06E-07	2.70E-07	

Note: Columns 2-5 show the posterior means for the parameters indicated for the controls by subgroup. Columns 6-9 show the posterior means for parameters from the treated sample. At bottom of the table the rest of the posterior mean parameters for controls and treated.

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